

WEEK-2

Malware detection using CNN

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2. Title: Malware detection using CNN

3. Objective / Problem Definition of the Project

The main problem addressed by this project is the increasing difficulty of detecting malicious software, especially new and modified malware variants. Traditional signature-based detection methods depend on previously known malware patterns and may fail when malware is altered or previously unseen. Therefore, there is a need for an effective automated approach that can identify malicious software accurately and distinguish it from legitimate software.

4. Datasets Available

Dataset Name	Purpose	Dataset URL
Malware Analysis Datasets: Raw PE as Image	Malware/goodware classification using image representations of PE files.	https://ieee-dataport.org/open-access/malware-analysis-datasets-raw-pe-image
EMBER	Static Windows PE malware detection using extracted PE features.	https://github.com/elastic/ember
MalVis	Malware classification using visual representations of executable files.	https://web.cs.hacettepe.edu.tr/~selman/malevis/

Selected dataset: Malware Analysis Datasets: Raw PE as Image

5. Detailed Description of the Selected Dataset

5.1 Dataset Name

Malware Analysis Datasets: Raw PE as Image

5.2 Dataset Author

Angelo Oliveira

5.3 Rows and Columns Count

Columns: 1,026 columns in the tabular representation — 1 hash column, 1,024 pixel columns (pix_0 to pix_1023), and 1 malware target column.

Rows: The exact number of rows should be taken from the complete dataset file being used by the team. This ensures that the number reported in the submission matches the actual copy of the dataset used in the experiment.

5.4 Rows and Columns Description

The 1,024 pixel values can be arranged into a 32×32 grayscale image because $32 \times 32 = 1,024$. Pixel values range from 0 to 255, where lower values represent darker shades and higher values represent lighter shades.

5.5 Target Label Description

The target column is named malware and contains two classes:

Label	Meaning
0	Goodware / benign software
1	Malware / malicious software

6. Experiment Setup

1. Data Loading: Load the dataset into Python using a suitable environment such as Jupyter Notebook or Google Colab.
2. Data Cleaning: Check the dataset for missing values, duplicate records, incorrect data types, and invalid pixel values.
3. Feature Selection: Use pix_0 to pix_1023 as the input features. The hash column is used only as an identifier and is not used as a CNN feature.
4. Target Selection: Use the malware column as the target variable.
5. Image Reconstruction: Reshape the 1,024 pixel values into a 32×32 grayscale image.
6. Normalization: Scale the pixel values from 0–255 to a 0–1 range to make CNN training more stable.
7. Dataset Split: Divide the data into training and testing sets. An 80:20 split can be used, with a validation set created from the training data if required.
8. CNN Model: Build a CNN with convolution, activation, pooling, flattening, and dense/output layers.
9. Training: Train the CNN using the training samples and their malware labels.
10. Evaluation: Evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix.

7. Inference / Justification About the Choice Made

The Raw PE as Image dataset was selected because it is directly related to cybersecurity and malware detection. It contains both malicious and benign executable samples, making it suitable for a binary classification problem.

The dataset is especially suitable for a CNN because the PE byte stream is represented as a 32×32 grayscale image. CNNs are designed to learn patterns from image-like data, so the 1,024 pixel values can be reconstructed into their original two-dimensional form and given to the network.

Another advantage is that the CNN can learn useful patterns automatically from the input representation rather than depending completely on manually selected PE features. This makes the dataset a good starting point for experimenting with deep learning techniques for malware detection.